

# OPPONG KWABENA ADUTWUM

## CONTACT

✉ oppongkwabena1777knust@gmail.com  
✉ kaoppng5@st.knust.edu.gh  
☎ +233 53 529 9655  
📍 Kumasi, Ghana  
➤ [linkedin.com/in/kwabena-oppng-adutwum](#)  
➤ [github.com/Oppong-pv](#)  
➤ [leetcode.com/u/Oppong-pv](#)  
➤ [oppng-pv.github.io](#)

## TECHNICAL SKILLS

### Programming Languages

- Python
- C++

### Data & Analytics

- NumPy
- SQL
- Jupyter Notebooks

### AI & ML Tools

- Scikit-learn
- Groq AI API
- Claude Code / Claude Cowork

### Web & Backend

- React Native / Expo

### Dev Tools & Design

- Git / GitHub
- VS Code
- Figma
- Linux (Kali)

### Core Knowledge

- Machine Learning Fundamentals
- PID Control / Robotics
- Data Structures & Algorithms
- OSI & TCP/IP Networking

## LANGUAGES

- English — Native
- French — Intermediate
- Twi — Native

## CERTIFICATIONS

- Hashgraph Developer Course — Hedera (Jan 2026)
- Hedera Certified Foundation (Apr 2026)
- IPPF Competitor — Brewer Foundation & NYU (2024-25)
- WASSCE General Science — Prempeh College (Sep 2025)

## GITHUB HIGHLIGHTS

- ml-specialization-coursera
- oppng-py.github.io
- leetcode-solutions

## INTERESTS

- Mathematics
- Machine Learning & AI
- Robotics & Embedded Systems
- UI/UX & Visual Design
- Movies & Video Games
- Sports (Football)

## MEMBERSHIPS

- Prempeh College Robotics Club (2023–2025)
- ACES, KNUST (2025–Present)
- AmaliTech Coding Club (2025–Present)
- Microsoft User Group — KNUST (2025–Present)
- Claude Builder Club — KNUST (2025–Present)

# ASPIRING MACHINE LEARNING ENGINEER

Applied ML · Robotics · Embedded Intelligence · Full-Stack Development

## PROFESSIONAL SUMMARY

First-year Computer Science student at KNUST with a strong interest in machine learning, robotics, and embedded systems. National Robotics Champion (Robofest Ghana 2025). Regional Cybersecurity Champion (2024). Currently studying Andrew Ng's ML Specialization (Stanford/Coursera) with focus on supervised learning, vectorization, and NumPy.

## EXPERIENCE

### Machine Learning Trainee — Zap Tech Technologies

Remote | May 2025

- Completed a 2-week introductory ML program covering data collection, preprocessing, model training, and evaluation.
- Cleaned datasets using Pandas and NumPy, handling missing values and feature scaling.
- Trained classification models with Scikit-learn and compared performance using accuracy, precision, recall, and F1-score.

### Online Safety Analyst & Volunteer — Solutions for Life Initiative Ghana (SFLIG)

Ghana | Oct 2022 – Present

- Joined as volunteer (Oct 2022) for community support projects; promoted to Online Safety Analyst (Aug 2023), managing social media and website with focus on internet security.
- Runs speaking engagements with girls from orphanages and rural communities on online safety topics.

### Co-Founder & Software Developer — Arethos Tech Initiatives

Kumasi, Ghana | 2026 – Present

- Co-founding and leading development of Arethos, a mental wellness app for KNUST students. Managing the technical roadmap, implementing AI-powered features, and coordinating a small dev team.
- Won 2nd Place at ACES CodeFest 2026 UI/UX Challenge. Redesigned the ACES KNUST website for mobile across all 9 pages as a team of 5, building a clickable Figma prototype.

## EDUCATION

### Kwame Nkrumah University of Science & Technology (KNUST)

B.Sc. Computer Science | Kumasi, Ghana | Expected 2029

- Relevant Coursework: Networking & Internet (OSI/TCP/IP, IPv4, subnetting), Calculus I, Probability & Statistics, C++ / CS & Math Programming

### Prempeh College (Senior High School)

WASSCE — General Science | Kumasi, Ghana | Graduated 2025

## PROJECTS

### Autonomous Line-Following Robot

LEGO MINDSTORM EV3 · PID Control Algorithm

- Designed PID control algorithm (a core ML precursor) for autonomous navigation; National Champion at Robofest Ghana 2025.
- Optimized sensor-motor feedback loops, directly applicable to reinforcement learning and robotics control.

### Class Attendance Monitoring System

React Native · Django · PostgreSQL · Bluetooth

- Architected biometric attendance platform serving 50 students in prototype testing, reducing manual roll-call by ~80%.
- Built Django REST API handling concurrent Bluetooth fingerprint scans, gaining experience in data ingestion at scale.

### Solar-Powered Automated Water Dispenser

Arduino · Ultrasonic Sensors · C++ · Solar

- Built a solar-powered water dispensing system using dual ultrasonic sensors to detect container placement and fill level, automatically controlling a solenoid valve.
- Implemented debounce logic, buzzer feedback, LED status indicators, and off-grid solar power for deployment without reliable electricity.

### Arethos — Mental Wellness App (In Development)

React Native · Expo SDK 54 · SQLite · Groq AI

- Cross-platform mental wellness app with AI-powered journaling, mood tracking, and adaptive reflections. Built with local-first storage and Groq AI integration for personalized responses.

## HACKATHONS & COMPETITIONS

### National Cybersecurity Challenge — Regional Champion (2024)

- Competed in a knowledge-based quiz testing cybersecurity fundamentals, threat analysis, and security best practices.

### Robofest Ghana — National Champion — Robotics (2025)

- Built autonomous robots under timed conditions to complete designated tasks; designed and assembled within strict time limits.

### ACES CodeFest 2026 — 2nd Place — UI/UX Design (July 2026)

- Redesigned the ACES KNUST website for mobile across all 9 pages as a team of 5. Built a high-fidelity clickable prototype in Figma with real-time collaboration, focusing on visual hierarchy, navigation, and user flows.

## LEADERSHIP & VOLUNTEERING

### Club Leader — Prempeh College Cybersecurity Club (2023–2025)

### Participant — International Public Policy Forum (IPPF) (2024)

References available upon request